

CSE 405: Computer Security

Class Test 1 – May 7 2025

Duration: 25 minutes

Total Marks: 20

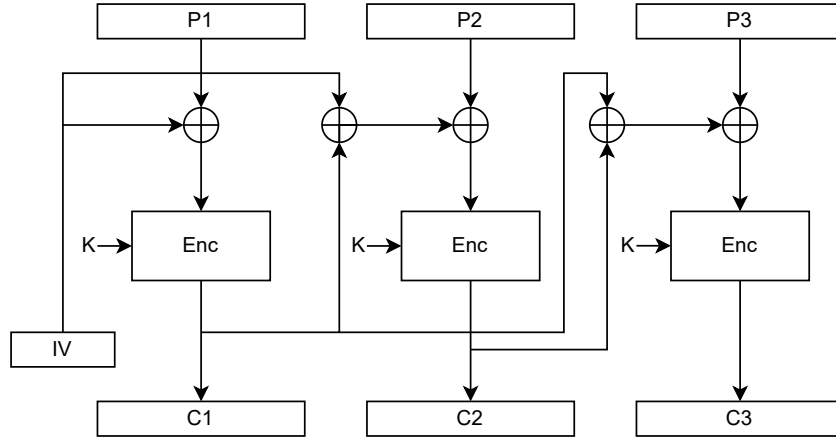
Name: Eve

ID: 666

Multiple Choice Questions

$6 \times 1 = 6$ marks

1. A file-sharing service lets users upload documents, but doesn't check file types. Attackers start uploading executable scripts disguised as documents. What principle was not followed?
 - A. Separation of responsibility
 - B. Security is economics
 - ✓C. Ensure complete mediation
 - C. Consider human factors
2. You have designed an encryption system, but if someone can access the source code, they can easily decrypt the messages. What principle is violated?
 - A. Fail-safe defaults
 - B. Design in security from the start
 - C. Security is economics
 - ✓D. Shannon's Maxim
3. Which of the following is essential for the key matrix in Hill cipher to be valid for encryption?
 - A. It must be symmetric
 - B. It must have only even elements
 - ✓C. It must be invertible modulo 26
 - C. It must be of size $n \times n$
4. An additive cipher can never be a polyalphabetic cipher.
 - ✓A. True
 - A. False
5. Which of the following keys is *invalid* for a multiplicative cipher over modulo 26?
 - A. 3
 - B. 5
 - ✓C. 13
 - D. 7
6. Which principle ensures that data cannot be altered without detection?
 - A. Confidentiality
 - B. Authentication
 - C. Availability
 - ✓D. Integrity



Consider the above block cipher encryption diagram. Write down the encryption equations for C_1 , C_2 , and C_3 . **3 marks**

$$\begin{aligned} C_1 &= E_k(P_1 \oplus IV) \\ C_2 &= E_k(P_2 \oplus C_1 \oplus IV) \\ C_3 &= E_k(P_3 \oplus C_2 \oplus C_1) \end{aligned}$$

Write down the decryption equations for P_1 , P_2 , and P_3 . **3 marks**

$$\begin{aligned} P_1 &= D_k(C_1) \oplus IV \\ P_2 &= D_k(C_2) \oplus C_1 \oplus IV \\ P_3 &= D_k(C_3) \oplus C_2 \oplus C_1 \end{aligned}$$

Which part(s) of the cryptographic process can be parallelized: encryption or decryption? **2 marks**

Answer: Only decryption can be parallelized. Encryption cannot be parallelized because each ciphertext block depends on the previous ciphertext block.

Suppose you are Mallory. You intercept the ciphertext (IV, C_1, C_2, C_3) and reorder it like this: (IV, C_2, C_1, C_3) . Determine what Bob would recover after decryption, expressed in terms of the original ciphertext blocks and the plaintext blocks. **6 marks**

$$\begin{aligned} P'_1 &= D_k(C_2) \oplus IV \\ P'_2 &= D_k(C_1) \oplus C_2 \oplus IV \\ P'_3 &= D_k(C_3) \oplus C_1 \oplus C_2 \end{aligned}$$

From the original decryption equations, we know:

$$\begin{aligned} D_k(C_1) &= P_1 \oplus IV \\ D_k(C_2) &= P_2 \oplus C_1 \oplus IV \\ D_k(C_3) &= P_3 \oplus C_2 \oplus C_1 \end{aligned}$$

Substituting these into the new equations gives:

$$\begin{aligned} P'_1 &= P_2 \oplus C_1 \\ P'_2 &= P_1 \oplus C_2 \\ P'_3 &= P_3 \end{aligned}$$